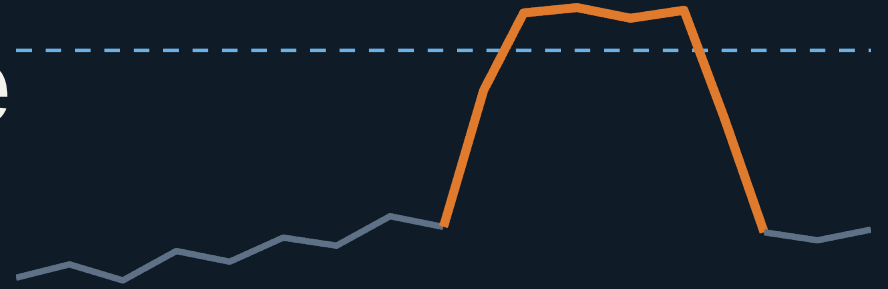


Bridging Network Telemetry and Generative AI With Elasticsearch and LLM



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Introduction

- Why “is it the network” hard to answer during an incident?
- Shape NetFlow data so an LLM understands it
- How function calls let an LLM answer in plain English

Disclaimer: The views and opinions expressed in this presentation are my own and do not represent those of my employer.

Agenda

- Domain Gap
- The Incident
- Netflow Basics
- Principle
- How it works
- Backend
- Trust
- Demo
- Takeaway

The Domain Gap

- Network engineers – show ip route
- SREs
 - kubectl get pods
 - GET _cluster/health
- Everyone else – “is it the network?”

The incident

- Routers already sends traffic records (NetFlow) to time series database
- IP ranges, BGP (Border Gateway Protocol) prefixes and link sizes only make sense with network knowledge
- The question bounces between teams until the network engineer shows up

Acme Ops

Channels

- # general
- # sre-oncall
- # inc-erp-slow**
- # network-ops
- # app-erp

Direct messages

- Priya (SRE)
- Marco (App)

inc-erp-slow
SEV-2 · ERP slow for HQ users · IC: Priya

M

Marco App team · 1:12 PM
ERP is timing out for users at HQ.
Is it the network?
+1 4

P

Priya SRE on call · 1:18 PM
Pods are healthy, CPU and memory normal.
Looks fine on our side.

D

Dana Database team · 1:25 PM
Query latency is normal.

M

Marco App team · 1:31 PM
Can anyone read the network data?
eyes 3

S

Sam Network engineer · 1:58 PM
Just joined. Pulling up NetFlow now...

Message #inc-erp-slow

Netflow Basics

- The access log of the network
- Metadata only: no payloads or packet contents
- Each record says who talked to whom, on which service, how much data, and when
- Routers send records continuously to a collector

What the router sends	What it means
Source IP 10.10.8.77	→ Who sent it: a laptop at HQ
Destination IP 10.30.2.10	→ Who received it: the file server
Port 445	→ Which service: Windows file sharing
Start / end 13:05 - 13:06	→ When it happened
Packets 850,000 (1.2 GB)	→ How much data moved
Interface Gi0/0/1	→ Which road: the link to the data center
Route 172.16.0.2	→ Next hop: the next router on the way
ToS / DSCP AF21	→ QoS: how the traffic is prioritized

Principle

- LLM doesn't read raw flow records
- It makes function calls that run Query DSL queries
- Elasticsearch executes the query against indices and data streams
- Why? LLMs are unreliable at math



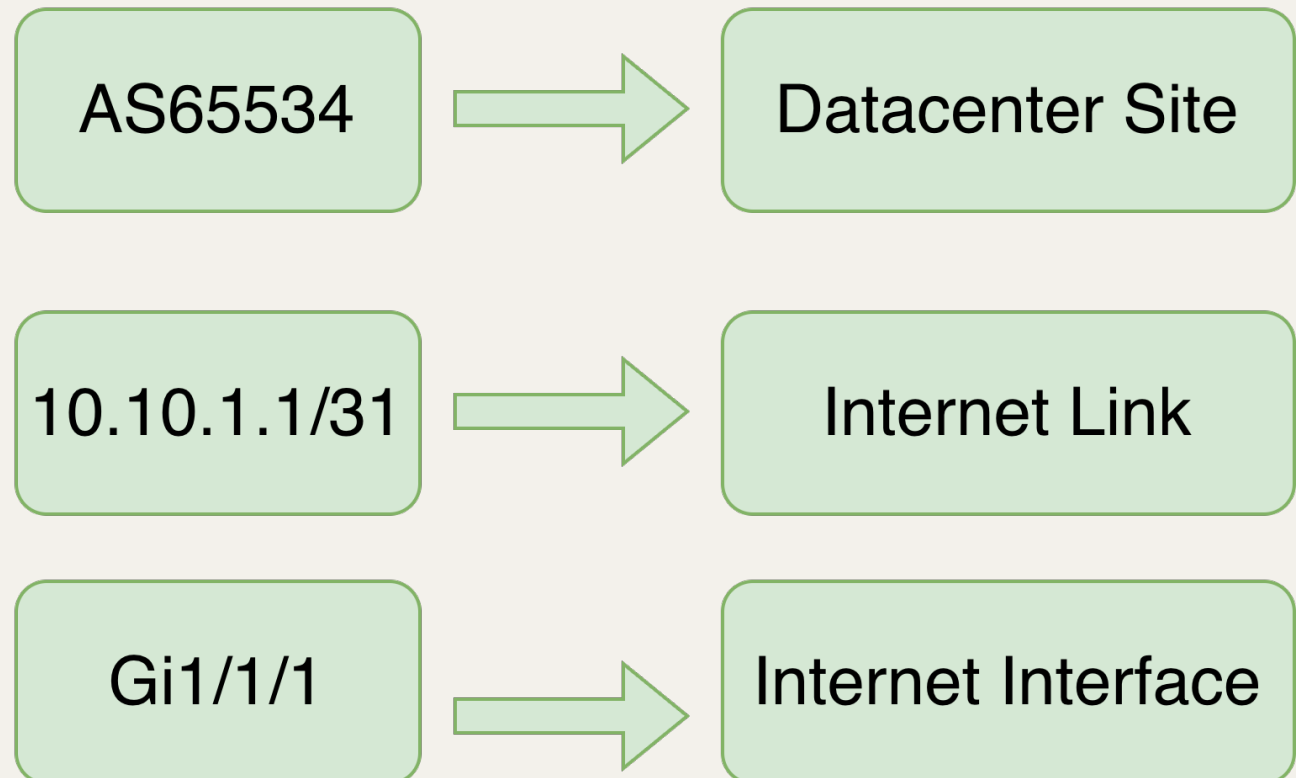
How it works



STEP 1

Make the data readable

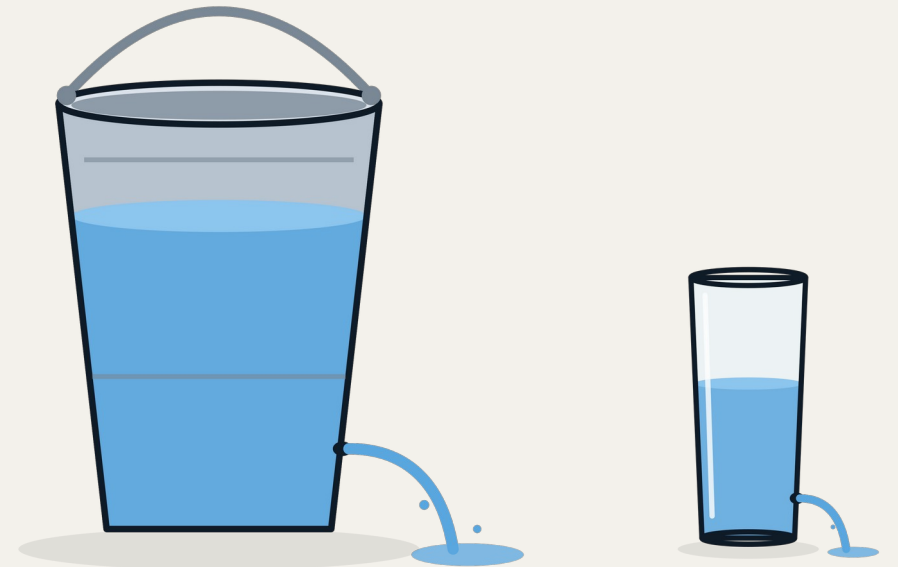
- Store names, not interface numbers
- IP Address – Point-to-point, server subnets
- ASN – Site



STEP 2

Add context

- Link size:
 - Internet link is 1 Gbps and 800 Mbps is 80% full
 - Data center link is 10 Gbps and 800 Mbps is 8% full
- Schedule
 - 12am – 5am: nightly backup
 - 9am – 5pm: business hours



STEP 3

Translate the question

1. Who is filling the internet link right now?
2. LLM fills in
 - **Tool:** top talkers
 - **Window:** last hour
 - **Link:** Gi0/0/0 (internet uplink)
 - **Rank by:** conversation
3. Elasticsearch question in JSON

QUESTION IN JSON

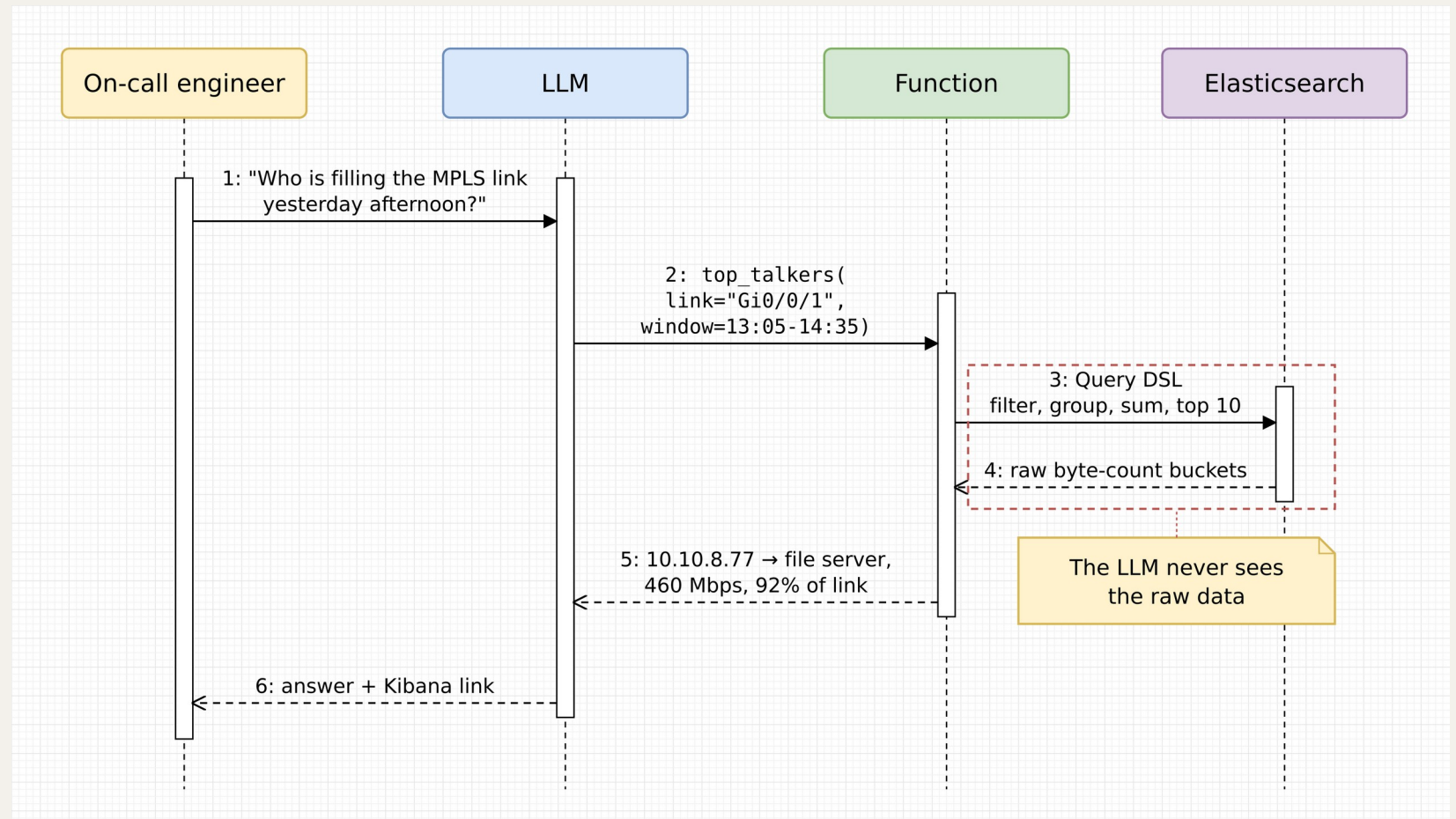
```
{
  "size": 0,
  "query": {
    "bool": {
      "filter": [
        { "term": { "interface.out.name": "Gi0/0/0" } } ],

```

STEP 4

Return the answer

- Code does the math: speeds, percentages, peaks and busy periods.
- LLMs are unreliable at arithmetic, so return the answer, not raw numbers.



Backend

Question	Function Call
Which link is it?	Network inventory
Was a link full?	Link utilization
Who was using it?	Top talkers
Is that normal?	Compare to baseline
Answer	Bottom line, evidence, next steps

Trust

- Every answer links to its evidence
- Links open the Kibana dashboard with the same filter and time window
- LLMs sometimes hallucinate, so checking takes one click

Demo

- Prompt:
 - Why is the GPU cluster so slow this morning?
 - What is using the internet right now? Is that Normal?

<https://github.com/penguin02007/FlowWatch>

Takeaway

- Enrich at ingest: names, not numbers
- Baseline matters more than the threshold
- Tell the model what time it is

Thank You

<https://github.com/penguin02007/FlowWatch>